A Paradoxical, Self-Referential System of Logics

Three non-classical logics as resolutions of one universal diagonal, with five machine-checked components

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Lean 4 witnesses: `github.com/inventor1975/ZTL` (`lean/ZTL.lean`, `lean/QuantumWitness.lean`, `lean/Contextuality.lean`, `lean/JunctionWitness.lean`)

Verify from zero: §6 — four files, four commands, no dependencies

Paradox and Truth — the progenitors of Logic!

Abstract

We propose a reading of three non-classical logics — intuitionistic / type-theoretic, quantum (orthomodular), and Zero-Trust Logic (ZTL) — as three disciplined responses to a single self-referential construction: the universal diagonal underlying the liar, Russell, Cantor, Gödel and Tarski (Lawvere 1969; Yanofsky 2003). Each retains all classical structural principles *save one*, and each thereby houses the diagonal differently *from within*; classical logic itself appears only as the baseline that retains everything and is therefore repaired from the metalanguage instead. Two features are invariant across the three: non-contradiction with its associated reductio (a retained *principle*), and the diagonal itself (a retained *obstruction*). We isolate the parts of this picture that are formal rather than expository — the complementary sacrifice made by ZTL and by orthomodular quantum logic, and the combinatorial core of quantum contextuality (Mermin–Peres, GHZ) and the junction theorem of composition — and record them as five machine-checked components in Lean 4 on the *empty axiom list* (no classical choice, no quotient, no propositional extensionality). A closing section, explicitly marked as a reading, draws the consequence of the contextuality witness: the quantum "corner" pays its laws only in the DESCRIPTIONS; at the level of states the diagonal's premise itself fails — the self-negating fixed point exists physically — so the paradox, for the world, dissolves rather than being tamed. We are explicit throughout about what is established (all three logics; the Lawvere–Yanofsky unification), what is the author's (ZTL; the arrangement; the five witnesses), and what is analogy carrying no proof (the "cycle" itself). This is a synthesis and a reading, not a theorem, a merger, a new foundation, or a new field.

Keywords. self-reference; Lawvere fixed point; diagonal argument; quantum logic; orthomodular lattice; three-valued logic; paracomplete logic; Zero-Trust Logic; universal logic; Lean 4; machine-checked proof.

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1 Scope

The claim of this note is deliberately narrow. It is *not* that a new logic or a new foundation has been discovered, nor that the three logics below are amalgamated into one calculus — they are provably incompatible in the precise sense of §5. The unification of the classical antinomies under a single fixed-point schema is due to Lawvere [8] and, in the concrete set-theoretic form used here, to Yanofsky [15]; quantum logic is Birkhoff–von Neumann [1]; the intuitionistic corner is standard; the "general theory of logical systems" already exists as a field (universal logic, abstract model theory [17]). The single component original to the author is **ZTL** [16], which appears here only as one corner. What this note adds is (i) the *arrangement* of the three as three resolutions of one diagonal, (ii) five machine-checked components (§6) — the two complementary poles, the impossibility of a deduction theorem in MO_2 for *any* implication, the combinatorial core of contextuality, and the junction theorem of composition — and (iii) a closing reading (§7), marked as such. Everything expository is marked as such; everything proved is in §6 and reproducible from zero.

2 The universal diagonal

The classical antinomies are one construction in several presentations. Lawvere's fixed-point theorem makes this exact.

Lemma 2.1 (Lawvere [8])**.** *In a cartesian closed category, if there is a point-surjective morphism $\varphi : A \to Y^A$, then every endomorphism $g : Y \to Y$ has a fixed point $y : 1 \to Y$ with $g \circ y = y$.*

Contrapositively: a *fixed-point-free* endomorphism of Y obstructs any point-surjection $A \to Y^A$. Instantiating Y and g:

- $Y = 2$, $g = \neg$ (no fixed point): no surjection $A \to 2^A$ — **Cantor**;
- the same schema, read in an internal truth object with a fixed-point-free negation, yields the **liar** and **Russell**;
- read in a provability / representation setting it yields **Gödel** and **Tarski**'s undefinability (Yanofsky [15], §§3–5).

The common core is *self-application against a fixed-point-free map*: an object that names its own maps, composed with a g that has no fixed point. Following Yanofsky, we call this schema **the diagonal**. It is *pre-classical*: it is not built by classical logic but is a construction that any logic expressive enough to talk about its own sentences must confront. A logic that ignores it does not escape it; it trivializes on it.

3 Three resolutions

A logic admitting self-reference must do something disciplined with the diagonal or collapse.

Classical logic is the baseline, not a resolution. It is stated here once and then set aside, because it is the zero against which the others are measured: it retains *every* structural principle and therefore has nothing to house the diagonal *with*. It explodes on the unlocalized liar and is repaired from the *metalanguage* — Tarski's hierarchy sits above the logic, not inside it. So its "resolution" is of a different kind from the three below, which each resolve the diagonal *within* themselves; and retaining everything is what it pays for in the hierarchy. Version 1.0.0 of this note listed classical logic as a fourth row, which counted it among the "non-classical logics" of its own subtitle and suggested a symmetry that does not hold: none of this note's five machine-checked components concerns the classical corner. It is a baseline here, and nothing about it is proved.

There are (at least) three disciplines that resolve the diagonal from inside, and each keeps all of the classical structural principles but one:

Logic	**Principle dropped**	**Resolution of the diagonal**
Intuitionistic / type-theoretic	excluded middle; unrestricted Type : Type	bans self-typing — the universe hierarchy; Girard's paradox is the diagonal of naïve type theory (Coquand [3]; Hurkens [4])
Quantum (orthomodular)	distributivity	the self-negating join lives as a genuine element; complementation stays total
ZTL	identity $p \to p$, excluded middle, double negation	pointwise quarantine: the liar has no fixed value; it is flagged, not exploded [16]

One honesty point: only the ZTL row is the author's; the other two are the standard content of their fields, curated here, not discovered.

4 Two invariants

Across the three, two things do not move — a dual pair, one positive and one negative.

The floor (a retained principle). Non-contradiction and the reductio it licenses, $(p \to \neg p) \to \neg p$, survive in every corner. In the quantum lattice $x \wedge x^{\perp} = 0$; in the ZTL matrix $\neg(p \wedge \neg p)$ is designated for every value; both are among the propositions of §6.

The ceiling (a retained obstruction). The diagonal survives in every corner, precisely because each logic is expressive enough to state it. What differs is only the *manner of housing* it (§3).

The two are linked: **non-contradiction is the guard against the diagonal.** A logic that surrendered non-contradiction would not tame the diagonal but be trivialized by it (*ex contradictione quodlibet*). Where the obstruction persists, the floor stands; that is why both are invariant.

5 No consistent amalgamation

Why an irreducible plurality rather than a hierarchy or a common refinement?

The three pairs fail to merge, but not all in the same way, and the difference is worth stating exactly.

Two pairs clash at the level of what is validated. Quantum logic retains $x \vee x^{\perp} = 1$; ZTL withholds $p \vee \neg p$ at the mark — no single valuation structure validates both a total complementation law and its pointwise failure. Type theory demands strong normalization (every term terminates); the ZTL solver houses non-termination as the transient phase N (§6).

The third pair — quantum and intuitionistic — is *not* incompatible in that sense, and it would be an overclaim to say so: intuitionistic logic does not negate excluded middle, it merely lacks it, and a common refinement of the two sets of retentions exists. But look at what it is. Distributivity (Heyting algebras are distributive) plus excluded middle, double negation and total complementation is a Boolean algebra: the join of the quantum and intuitionistic retentions is *classical logic itself.* The pair can merge — and merging expels them from the family, because the merge is exactly the baseline of §3, which resolves nothing from within: it explodes on the raw liar and is repaired from the metalanguage. So the three are incomparable *as resolutions*: two pairs cannot share a valuation structure, and the third pair can share one only by ceasing to resolve.

Version 1.0.0 ran this argument through the baseline in a muddled form — "any system retaining all four families would validate distributivity and its negation, hence be trivial; that system is classical logic" — muddled because classical logic is not trivial. But the old argument was gesturing at something true, and the repair is to state it, not to delete it: the baseline is where the mergeable pair lands. The informal picture of an *orbit* around an unreachable centre is gone; it was marked as metaphor and did no work.

What remains is the content: the pairwise incompatibility above and the concrete complementarity of §6. No claim of a formal cyclic structure is made.

6 The ZTL–quantum mirror, machine-checked

The one formalizable part of the picture is the *complementary sacrifice* made by ZTL and by orthomodular logic. Both are recorded below as propositions verified in Lean 4 over finite carriers by `decide`, on the **empty axiom list**, in bare Lean (no `mathlib`, no imports).

Pole A (ZTL). Let $M = \{\mathsf{T}, \mathsf{F}, \mathsf{Z}\}$, with connectives generated by the *zero-trust lift*: for a Boolean f, $\mathrm{lift}(f)(\vec{x}) = \mathsf{T}$ iff f returns `true` under every classical reading of the arguments (Z read as both `true` and `false`), else F. Then in M:

Proposition 6.1 (`lean/ZTL.lean`, empty axiom list)**.**

(i) non-contradiction — $\neg(p \wedge \neg p) = \mathsf{T}$ *for all* p*;*

(ii) distributivity in both directions — $p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$ *and dually, for all* p, q, r*;*

(iii) failure of identity, excluded middle and double negation — $p \to p$, $p \vee \neg p$, $\neg\neg p \to p$ *are not designated for all* p *(each is falsified at* $p = \mathsf{Z}$*).*

Here the third symbol Z is a **status mark, not a truth value**: verdicts are two-valued, Z marking an unverified input (greedy register) or, reused positionally in the solver, the transient phase N = "not yet" (Kleene's undefined), which never labels an input. ZTL is thus **paracomplete** (truth-value gaps, no gluts), not paraconsistent [16].

Pole B (orthomodular). Let MO_2 be the smallest non-distributive orthomodular lattice: $\{0, 1\}$ together with two complementary pairs $(a, a^\perp), (b, b^\perp)$, all four proper elements mutually incomparable, with $x \wedge x^\perp = 0$, $x \vee x^\perp = 1$. Then:

Proposition 6.2 (`lean/QuantumWitness.lean`, empty axiom list)**.**

(i) non-contradiction — $x \wedge x^\perp = 0$ *for all* x*;*

(ii) excluded middle and double negation — $x \vee x^\perp = 1$ *and* $x^{\perp\perp} = x$ *for all* x*;*

(iii) failure of distributivity — $a \wedge (b \vee b^\perp) = a \wedge 1 = a$*, whereas* $(a \wedge b) \vee (a \wedge b^\perp) = 0 \vee 0 = 0$*.*

The quantum surrender is necessary, not a choice of arrow. An ortholattice has no canonical implication, so one might suspect that whatever MO_2 loses is an artefact of the connective chosen for it. It is not. The deduction theorem

$$a \le (b \to c) \quad \Longleftrightarrow \quad a \wedge b \le c$$

says that $b \to c$ must *be* the greatest element whose meet with b lies below c — a relative pseudo-complement. A lattice possessing one for every pair is a Heyting algebra, hence distributive; MO_2 is not.

Proposition 6.3 (`lean/QuantumWitness.lean`, empty axiom list)**.**

(i) modus ponens survives *the Sasaki hook* $x \to_s y := x^\perp \vee (x \wedge y)$*:* $x \wedge (x \to_s y) \le y$ *for all* x, y*;*

(ii) no implication whatever gives MO_2 the deduction theorem — *there is no binary operation* $\to$ *on* MO_2 *with* $a \le (b \to c) \Leftrightarrow a \wedge b \le c$ *for all* a, b, c*. Witness: at* $b = a$*,* $c = 0$ *the set* $\{x : x \wedge a \le 0\}$ *contains the mutually incomparable* $a^\perp, b, b^\perp$*, so any such arrow would have to place* $a \to 0$ *above their join* 1*, while* $1 \wedge a = a \not\le 0$*.*

This is the difference between measuring a choice and stating a fact about the object: not "MO_2 under the Sasaki hook lacks the deduction theorem", but "MO_2 cannot have one".

The mirror. ZTL retains distributivity and surrenders excluded middle, double negation and identity; orthomodular logic retains excluded middle and double negation and surrenders distributivity; both retain non-contradiction. Informally: ZTL fails where a proposition meets *itself*; quantum logic fails where propositions *combine*. Proposition 6.3 sharpens that second clause into a necessity: what MO_2 cannot do is *discharge a premise while a context still stands*, and no choice of arrow repairs it. The informal reading and the machine-checked obstruction agree on the same locus.

Remark (the honest mark)*.* The two "retentions of excluded middle" are not instances of one theorem in one system. In MO_2, $x \vee x^\perp = 1$ is a *lattice identity* and $x^{\perp\perp} = x$ holds because the orthocomplement is involutive *by the choice of an ortholattice* — mild, structural. In M, the *failure* of $p \vee \neg p$ is a computation in a *logical matrix*, where validity means designation. These are facts in two different formalisms that rhyme; the "mirror" is an analogy made precise on each side, not a single cross-formalism theorem. This is the boundary between what is proved (each proposition) and what is read (their pairing).

Indeed the mirror provably does *not* lift to a lattice duality, and the obstruction is instructive. A duality functor would have to intertwine the two negations; but they are of different species. The orthocomplement of MO_2 is *involutive* ($x^{\perp\perp} = x$), while the ZTL negation is *not* ($\neg\neg\mathsf{Z} = \mathsf{T} \ne \mathsf{Z}$; double negation is one of the laws ZTL surrenders). No functor matches a non-involution to an involution — so the very asymmetry that *makes* the mirror (DNE, kept by one side and dropped by the other) is what *forbids* it being a duality. The point locates ZTL precisely: its mark descends from Łukasiewicz's $Ł_3$ [9], whose negation is involutive, and the known bridge from orthomodular quantum logic to many-valued logic runs through infinite-valued Łukasiewicz logic (every Birkhoff–von Neumann logic is a partial such logic [14]), whose negation is likewise involutive. ZTL is the paracomplete relative that *breaks* the involution its own ancestor and its quantum cousin both keep; its originality and the non-duality of the mirror are one and the same fact.

The pair extension. On TWO quanta the mirror sharpens. At the singlet state, pair propositions are true ("spins are opposite", "total spin = 0") while *every* local proposition — either particle, any axis — has probability $\frac{1}{2}$: a true join whose joinees are all unverified. The law that falls is the covering law $a \vee b$ true $\iff$ a true or b true — a law the ZTL matrix keeps as a theorem (`cover_or_T`). Lattice expressibility, by contrast, does NOT fall: the singlet line is reachable from local propositions by meet and join (measured in the repository stand `dilemmas/quantum_pair.py`); what is not compositional is the truth VALUE, not the lattice. The seam itself is kernel-checked (`lean/JunctionWitness.lean`, empty axiom list, exact integer arithmetic): the *junction theorem* — the singlet lies in the join of the two product atoms yet in neither atom and in no local plane of either factor. Plugging two quantum logics together makes the COVER face of the disjunction (one face of the trio ZTL surrenders) fall for quantum at the seam of composition, in correlation form — entanglement as a lattice fact; the LEM identity $x \vee x^{\perp} = 1$ itself stays intact at every size. The falls saturate along the ladder (repository stand `dilemmas/quantum_ladder.py`): one quantum costs distributivity (and modus ponens for the material arrow; the Sasaki arrow keeps both identity and modus ponens), two quanta cost the valuation itself, and beyond two nothing new falls, while $\neg\neg$, non-contradiction and the LEM-identity hold at every size.

Proposition 6.4 (`lean/Contextuality.lean`, empty axiom list)**.**

(i) Mermin–Peres*: no assignment of* ± 1 *to the nine cells of the magic square satisfies the six sign constraints (rows* $+1$*; columns* $+1$*,* $+1$*,* -1*) —* 0 *of* 512*, by kernel enumeration;*

(ii) GHZ*: no assignment of* ± 1 *to the six single-particle values satisfies* $XXX = +1$*,* $XYY = YXY = YYX = -1$ *—* 0 *of* 64*.*

The honest split: Lean proves the COMBINATORIAL impossibility for these sign patterns; that the patterns are quantum's is direct operator arithmetic, verified in the repository stand. Together with the pair measurement: the local truth-value addresses are not merely empty — no bivalent valuation exists to fill them. The quantum "Z" is an ontic vacancy, not an epistemic flag; this is the precise boundary of the kinship between superposition and ZTL's mark.

Reproduction (Lean 4.29.1, measured 2026-07-18/19):

```
$ lean lean/ZTL.lean             # 13 objects, no axioms,  ~0.3 s
$ lean lean/QuantumWitness.lean  # 11 objects, no axioms,  ~0.2 s
$ lean lean/Contextuality.lean   #  3 objects, no axioms,  ~0.3 s
$ lean lean/JunctionWitness.lean #  8 objects, no axioms,  ~0.2 s
```

Every `#print axioms` line reports *"does not depend on any axioms"*, definitions included: no `Classical.choice`, no `Quot.sound`, no `propext`. The intuitionistic and classical corners are, by contrast, *metatheorems* (underivability of excluded middle; failure of strong normalization for Type : Type; "classical as centre") and are not formalized here by design; the diagonal itself, ranging over three logics that are not objects of one theory, is likewise not a single formal object.

7 The world is not a corner (a reading)

Everything in this section is interpretation, anchored on the measured cells of §6; no formal claim is made beyond Propositions A–C.

The three corners of §3 are DESCRIPTIONS, and it is the descriptions that pay the laws: distributivity, the covering law, the valuation. What Proposition 6.4 exhibits is not a broken inference inside any lattice — the lattice is consistent — but the absence of a semantic map: the world declines to be a classical model. The discrepancy is located at the INTERFACE between a logic and its world, and it is closed to every repair: no further verification, no better translation, no finer instrument removes it, because there is no valuation to recover (0 of 512). Where a discrepancy between derivation and measurement is normally an incident to be triaged, here it is a theorem.

The same relocation dissolves the diagonal. Lawvere's construction needs a fixed-point-free map; for every description's negation this premise holds — the ZTL tables prove $\neg v \neq v$ for all v, and a subspace never equals its orthocomplement, since no nonzero vector is orthogonal to itself — and so every description must TAME the resulting monster, each paying its price: the classical hierarchy of metalanguages, the type-theoretic tower of universes, ZTL's completeness tax for the quarantine. But at the level of STATES the premise fails: the flip has a fixed point, and it is physical — $X|+\rangle = |+\rangle$, a state prepared and measured on hardware (repository record, IBM Heron, 2026-07-17). The two negations must not be conflated — the lattice orthocomplement remains fixed-point-free while the state-level flip is not — and precisely this non-conflation is the point: the self-negating fixed point, a paradox in every description, EXISTS in the world as a citizen, and its existence costs the world nothing — no hierarchy, no tower, no tax. The paradox is a disease of descriptions, not of the world.

Read this way, the cycle closes cleanly: the world supplies the obstruction (the interface discrepancy, permanent by Proposition 6.4) and natively houses the fixed point; the three logics are three ways of living with that supply, each paying exactly one law for its manner of description. The universe is not a fifth corner. It is what the corners are about.

8 Provenance

	Content	Status
Established	Lawvere's fixed-point theorem and the diagonal unification (Lawvere [8]; Yanofsky [15]); quantum logic and non-distributivity (Birkhoff–von Neumann [1]); contextuality (Kochen–Specker [7]; Mermin [10]; Peres [13]); intuitionism, the universe hierarchy, Girard's paradox (Martin-Löf [11]; Coquand [3]; Hurkens [4]); universal logic as a field (Béziau [17])	literature
Author's	ZTL, the zero-trust paracomplete logic [16]; the arrangement of the three as resolutions of one diagonal; the five Lean witnesses on the empty axiom list (the two poles, the impossibility of a deduction theorem in MO_2, the contextuality core, the junction theorem); the measured ladder of falls and its saturation	synthesis + machine-checked
Analogy (no proof)	the "cycle / orbit / centre"; "the floor guards the ceiling"; "each logic pays one principle so none pays the whole diagonal"; §7 entire — "the world is not a corner; the paradox is a disease of descriptions"	structural metaphor, marked

The defensible claim is narrow: three established non-classical logics align as three disciplined resolutions of one pre-classical diagonal, and the sharpest pair among them — ZTL and orthomodular logic — make complementary, machine-checkable sacrifices. The organizing picture is offered as a reading; its five machine-checked components — the two poles, the impossibility of a deduction theorem in MO_2, the contextuality core, and the junction theorem — are offered as proof.

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AI disclosure. This note was written in dialogue with the AI system Claude (Anthropic); all design decisions, the framing and final responsibility are the human author's. The reliability of §6 depends on neither the author nor the AI: both Lean files verify against the Lean 4 kernel on the empty axiom list. Everything outside §6 is offered as a reading and marked as such.